

SPL 3 - Polymer Processing: MCQs

Q1. Which of the following processes is best suited for producing hollow thermoplastic parts like water tanks and playground equipment?

- A) Compression Moulding
- B) Injection Moulding
- C) Rotational Moulding
- D) Vacuum Forming

Q2. Which polymer processing method involves melting pellets, injecting them into a cold mould, and solidifying by cooling?

- A) Compression Moulding
- B) Extrusion
- C) Injection Moulding
- D) Blow Moulding

Q3. What is the key difference between thermoplastics and thermosets?

- A) Thermoplastics cure chemically, thermosets melt and cool repeatedly
- B) Thermoplastics can be remelted, thermosets cannot
- C) Thermoplastics are brittle, thermosets are ductile
- D) Thermosets are supplied as pellets, thermoplastics as liquids

Q4. Which process is most suitable for creating thin plastic bags and packaging films?

- A) Blow Moulding
- B) Compression Moulding
- C) Blown Film Extrusion
- D) Vacuum Forming

Q5. Which process uses compressed air to push molten polymer against the mould walls, creating a part with a hollow interior?

- A) Gas-Assisted Injection Moulding
- B) Extrusion
- C) Vacuum Forming
- D) Transfer Moulding

Q6. Which shaping process is most appropriate for thermosetting polymers?

- A) Blow Moulding

B) Compression Moulding

C) Injection Moulding

D) Extrusion

Q7. Vacuum forming is best suited for which type of parts?

A) Solid metal replacement parts

B) Hollow, highly detailed internal components

C) Thin, shallow plastic trays and packaging

D) Complex parts with internal ribs and fine surface texture

Answer Key

Q1: C

Q2: C

Q3: B

Q4: C

Q5: A

Q6: B

Q7: C